

SMART INDIA HACKATHON 2026

AURA SENTINEL

AI-Powered Continuous Cyber Risk Quantification & Investment Optimization Platform

- **Problem Statement ID:** SIH26105 | Ministry of Education's Innovation Cell (MIC) & AICTE Cyber Security Cell
- **Theme & Category:** Blockchain & Cybersecurity | Software Solution (Enterprise & Campus Defense)
- **Team Name & ID:** ByteForce_1 (Official SIH Team ID: 180219)
- **Institute Name:** Skyline Institute of Engineering and Technology, Greater Noida
- **Team Leader & Members:** Aryan Baghel (Lead), Kirti, Hitesh Chauhan, Avinav Jha, Sharim Khan, Aman Sekh
- **Core Mission:** From abstract vulnerability ratings to actuarial rupee loss estimates — sealed in a tamper-evident audit ledger.

1. HOST TELEMETRY

Active Sockets & Compute Stress

2. RISK SCORING

Weighted Multi-Factor Exposure

3. FINANCIAL EXPOSURE

Prototype EAL (₹ INR) Model

4. SECURITY INVESTMENT

Pareto Knapsack Optimizer

5. AUDIT TRUST

SHA-256 Merkle Ledger

Slide 02 | The Real-World Cyber Risk Problem & AURA Solution

THE REAL-WORLD CYBER RISK PROBLEM

- ❌ **1. Static Risk Assessments:** Periodic questionnaires become obsolete as system conditions, active ports, and network services change continuously.
- ❌ **2. Technical Risk ≠ Business Impact:** Vulnerability ratings like 'High / Medium / Low' fail to communicate monetary balance-sheet exposure to executive boards and CFOs.
- ❌ **3. Security Budget Uncertainty:** Organizations lack a mathematically justifiable method to evaluate and prioritize candidate security controls against a finite budget.
- ❌ **4. Audit Log Integrity Vulnerabilities:** Traditional database audit logs remain vulnerable to insider modification, violating strict regulatory non-repudiation mandates.

AURA SENTINEL: CONTINUOUS DECISION ENGINE

- **Real-Time Telemetry:** Monitors active listening sockets, CPU/RAM stress, and sensitive credential exposures.
- **Continuous Risk Engine:** Calculates a normalized 0–100 multi-factor exposure score dynamically.
- **Financial Risk Model (Prototype):** Translates host stress into estimated operational downtime & Expected Annual Loss in ₹ INR.
- **What-If Scenario Simulation:** Empowers operators to model parameter changes & simulated MITRE threat mitigations.
- **Budget Knapsack Optimizer (Prototype):** Ranks security defense packages by ROSI (%) under explicit budget limits (e.g. ₹1 Crore).
- **Tamper-Evident Audit Ledger:** Anchors assessments and quarantine records in a SHA-256 Merkle blockchain.

PARADIGM SHIFT: FROM ABSTRACT CVE RATINGS → TO MEASURABLE, ACTUARIAL & ACTIONABLE DECISIONS

"From technical security signals to financially informed security decisions — backed by a tamper-evident audit ledger."
(Note: Financial exposure calculations operate as prototype models designed for institutional decision-support).

Slide 03 | Technical Approach: Architecture & Data Pipeline

1. REAL-TIME SYSTEM INPUTS

CPU • RAM • Disk • Processes • Ports • Sockets • Sensitive Files (psutil)

2. ATTACK SURFACE ENGINE

Listening Socket Enumeration • Port Classification • Active Process Remediation

3. CONTINUOUS RISK ENGINE

Multi-Factor Weighted Scoring Model (0–100 Normalized Exposure Metric)

4. FINANCIAL RISK MODEL [PROTOTYPE]

Downtime Cost Rate • Hourly Disruption • Prototype Expected Annual Loss (₹ INR)

5. WHAT-IF SIMULATION ENGINE

Interactive Slider Parameter Tuning • MITRE ATT&CK Wargame Simulation

6. INVESTMENT OPTIMIZER [PROTOTYPE]

Institutional Budget Constraint (₹) → Pareto Knapsack Optimizer → ROSI %

7. TAMPER-EVIDENT AUDIT LEDGER

SHA-256 Merkle Root Sealing • Proof-of-Work Mining • Integrity Verification

TECHNOLOGY STACK & STATUS

FRONTEND SOC LAYER: React 19 + Vite 8 + Cyber Audio SOC

Status: ● IMPLEMENTED

BACKEND MICRO-KERNEL: Python 3.10+ & FastAPI Async Kernel

Status: ● IMPLEMENTED

HOST TELEMETRY INGESTION: psutil Real-Time Socket & Metric Inspector

Status: ● IMPLEMENTED

AI / COMPUTER VISION: Ultralytics YOLOv8n + MediaPipe 0.10

Status: ● IMPLEMENTED

LOCAL STORAGE HUB: SQLite 3 with Write-Ahead Logging (WAL)

Status: ● IMPLEMENTED

CRYPTOGRAPHIC AUDIT: SHA-256 Merkle Ledger + PoW Demo

Status: ● IMPLEMENTED

PROTOTYPE EXPOSURE & ROSI: Actuarial FAIR Formulae + Knapsack

Status: ○ PROTOTYPE

ADVANCED RISK MODELING: Probabilistic Monte Carlo VaR (95%)

Status: ○ FUTURE

Slide 04 | Feasibility, Viability & Deployment Scaling

TECHNICAL FEASIBILITY & READINESS

- ✓ **Local-First Architecture:** Runs efficiently on standard quad-core CPUs (<1% host compute overhead) without mandatory cloud GPU clusters.
- ✓ **Zero Recurring License Fees:** Operates without paid third-party API dependencies, offering a viable alternative to costly proprietary suites.
- ✓ **Tested Standalone MVP:** Core host telemetry, port classification, What-If wargaming, and tamper-evident ledger are fully functional and demonstrable today.
- ✓ **Air-Gapped Operational Safety:** Can operate entirely inside closed institutional LANs with zero external telemetry data leakage.

- **IMPLEMENTED:** Host Telemetry, Attack Surface, Risk Scorer, What-If, Audit Ledger, Auto-Lock

- **PROTOTYPE:** Expected Annual Loss (EAL in ₹), Knapsack Investment Optimizer

- **FUTURE:** Monte Carlo VaR Distributions, Enterprise Forwarders, Multi-Institution Network

VIABILITY & DEPLOYMENT SCALING JOURNEY

PHASE 1: LOCAL MVP (Host Agent)

Standalone single-endpoint agent validating continuous telemetry & exposure.

PHASE 2: INSTITUTIONAL PILOT (Campus LAN)

Lightweight agents forwarding host metrics across internal campus nodes.

PHASE 3: ENTERPRISE INTEGRATIONS

Native connectors for Syslog, Fluentbit, and Wazuh log pipelines.

PHASE 4: MULTI-INSTITUTION CONSORTIUM

Decentralized consortium nodes sharing unforgeable cross-campus risk posture.

- **Modular Decoupled Design:** Asynchronous micro-services allow independent horizontal scaling.
- **Transparent Decision Support:** Baseline hourly rates and asset weights are fully configurable by institutional CISOs.

ARCHITECTURAL POSITIONING & DEFENSIVE REASONING

"AURA Sentinel is engineered as an actuarial decision-support and audit verification platform — it complements, rather than replaces, dedicated perimeter firewalls and endpoint detection infrastructure."

Slide 05 | Social, Economic & Institutional Impact

OPERATIONAL PARADIGM SHIFT

BEFORE AURA: [Raw Host Logs] → [Isolated Alert Spam] → [Qualitative CVE Tags] → [Suboptimal Budget Allocation]

AFTER AURA: [Live Telemetry] → [Continuous Risk Scoring] → [Actuarial Rupee Exposure] → [Knapsack ROSI Optimizer] → [Auditable Ledger]

SECURITY OPERATIONS TEAMS

- Continuous host attack-surface visibility
- Automated listening socket threat mapping
- Proactive MITRE wargame threat simulation
- Rapid risk-informed vulnerability remediation

MANAGEMENT & EXECUTIVE BOARD

- Actuarial financial visibility into cyber exposure (₹)
- Budget-constrained defense capital planning
- Transparent Return on Security Investment (ROSI %)
- Clear risk vs. expenditure trade-off visualization

AUDIT & COMPLIANCE OFFICERS

- Tamper-evident SHA-256 Merkle audit trails
- Indisputable cryptographic non-repudiation
- Single-click printable CISO Audit PDF report
- Meets CERT-In 180-day unalterable log mandates

ACADEMIC & ENTERPRISE INSTITUTION

- Eliminates wasteful security tool purchases
- Reduces costly operational downtime hours
- Local-first deployment without SaaS license fees
- Strengthens overall cyber resilience posture

CORE VALUE CHAIN: TECHNICAL SIGNALS → CYBER RISK → BUSINESS IMPACT (₹) → SECURITY INVESTMENT → AUDITABLE PROOF

Slide 06 | Research, Standards Alignment & Project Resources

NIST CSF 2.0

Core functions: Identify, Protect, Detect, Respond, Recover

MITRE ATT&CK v15

Wargame mappings: T1110, T1499, and T1046 recon

ISO/IEC 27001

Section A.12 operations security & logging alignment

CIS Controls v8

Controls 1 (Assets), 4 (Config), and 8 (Audit Logs)

CERT-In Directions

180-day unalterable incident log preservation mandate

DPDP Act 2023

Credential & PII scanning for unmasked Aadhaar/PAN

RBI / SEBI Guidance

Tiered risk threshold alerting & resilience reporting

The Open FAIR Standard

Actuarial cyber risk economics & loss magnitude

PROJECT REPOSITORY & VEF

GitHub: <https://github.com/aryanbaghel756-spec/AURA>

- Complete reproducible codebase & test suite
- 1-Click launcher scripts (run_aura_mvp.bat)
- Fully runnable local-first MVP build
- Evaluator documentation & live demo guide

Scan the QR code to inspect the live codebase:



CORE DIFFERENTIATOR & LIVE EVALUATION FLOW

TRADITIONAL SCANNERS: Detect → Alert

AURA SENTINEL: Detect → Quantify (₹) → Simulate → Optimize (ROSI) → Audit

5-MINUTE RUNNABLE DEMO PLAYBOOK:

- 0:00–0:45 | Module 01: Host Telemetry & Listening Sockets
- 0:45–1:30 | Module 02: Attack Surface & Port Threat Classification
- 1:30–2:15 | Module 04: Prototype Financial Exposure (EAL in ₹ INR)
- 2:15–3:15 | Module 05: What-If Slider Tuning & MITRE Wargame Simulation
- 3:15–4:15 | Module 06: Budget-Constrained Knapsack ROSI Optimizer
- 4:15–4:45 | Module 10: Cryptographic Blockchain Tamper Showdown
- 4:45–5:00 | Navbar: 1-Click Executive Printable CISO Audit Report